CAPSTONE TECHNICAL PROPOSAL

Care2ClaimZA-LLM: Fine-Tuned Gemma 3-1B for South African Healthcare Claims Adjudication & Triage

AI Research Foundations for Higher Education · Capstone Project Hackathon

Cohort	Pretoria Cohort
Submitted by	Mr. Clayton Baker, AI Champion, University of the Western Cape
Team	Deeply Baked
Pathway	Combine two: Conversational-classification hybrid assistant
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Elevator pitch

> **We are building a hybrid AI healthcare decision guide for South Africans experiencing distress so that they can navigate care and coverage with dignity.**

1. The problem

What problem are you solving, and for whom?

South African healthcare navigation suffers from opaque medical scheme rules, complex benefit brochures, rigid search tools, and misleading self-diagnosis engines. Consequently, patients face heightened health anxiety, unexpected out-of-pocket costs from misaligned network providers, and communication gaps with doctors. Care2ClaimZA-LLM addresses these challenges through two core capabilities. First, it automates administrative policy adjudication into In-Cover, Out-of-Cover, or Requires Pre-Authorisation categories. Second, it converts everyday layperson symptom descriptions into transparent, non-diagnostic policy guidance and triage advice aligned with statutory regulations.

The solution serves South African healthcare consumers and medical scheme members who require clear coverage answers without financial surprises. Additionally, it assists healthcare providers by streamlining clinical intake and reducing administrative delays.

Why does it matter in your context?

In the South African context, private healthcare navigation is governed by complex statutory regulations, including Prescribed Minimum Benefits and Designated Service Provider networks. However, severe information asymmetry exists between medical schemes and healthcare consumers. Convoluted policy rulebooks and dense clinical terminology frequently obscure

statutory rights. Consequently, patients face unexpected out-of-pocket medical debt, administrative claim rejections, and heightened health anxiety. Furthermore, language barriers in diverse communities make it difficult for individuals to articulate medical needs using formal scheme vocabulary. Care2ClaimZA-LLM addresses these systemic challenges directly. By translating everyday layperson symptom descriptions into transparent administrative guidance, the model demystifies statutory coverage rules and clarifies benefit entitlements. Ultimately, this safeguards patient dignity, mitigates unexpected financial harm, and promotes equitable healthcare access across South Africa while adhering strictly to local privacy legislation like the Protection of Personal Information Act.

What is out of scope?

Care2ClaimZA-LLM does not provide autonomous clinical diagnoses or prescribe medical treatments. Instead, it strictly delivers non-diagnostic administrative triage and benefit explanations. Additionally, calculating exact monetary out-of-pocket costs or procedure pricing remains outside the system scope. Finally, the model does not access private health records or personal health information. It relies exclusively on de-identified data and public regulatory frameworks to maintain strict privacy compliance.

2. Pathway

Pathway

Combine two

Which two pathways?

Conversational-classification hybrid assistant

One example

Input	Output
[Step 1: Conversational Intake & Symptom Triage] Molo doc, eish my body is hurting. I'm experiencing abdominal pain. [Step 2: Policy Adjudication & Claim Classification] Classify the following medical claim into one of the policy decision categories: 'In-Cover', 'Out-of-Cover', or 'Requires Pre-Auth'. Medical Specialty: General Medicine Clinical Description: This 68-year-old man presents to the emergency department for three days of cough, claims that he has brought up some green and grayish sputum. He says he does not feel short of breath. He denies any fever or chills.	Output: [Step 1 Output] I'm sorry to hear that. How long have you had this abdominal pain, and are you noticing other symptoms like blood in the stool? [Step 2 Output] In-Cover

3. Data plan

Type of data

Text

Where will it come from?

An existing dataset, We will generate synthetic data

Dataset name or link

https://huggingface.co/datasets/NickyNicky/medical_mtsamples and
https://www.kaggle.com/datasets/rudravpatel/healthcare-dataset-for-chatbot

Language(s)

South African English

How many examples will you aim for?

98

Who might be missing or under-represented?

Uninsured and unemployed populations relying solely on public healthcare are excluded, as the dataset focuses on private medical scheme rules. Additionally, non-English speakers and individuals with literacy constraints or cognitive impairments may be underrepresented due to English clinical corpora limitations. Finally, members of smaller, specialised medical schemes or niche plan tiers are omitted because their specific coverage exemptions differ from major market standards.

Licence and consent

Confirmed

4. Model

What kind of model will you use?

The system fine-tunes Google's Gemma 3 1B foundation model using Parameter-Efficient Fine-Tuning with LoRA.To ensure memory stability on an NVIDIA T4 GPU with 15 GB VRAM, the execution environment adopts a mixed_bfloat16 precision policy. LoRA backbone adaptation is applied at rank 8 with alpha equal to 16, modifying 1,304,576 trainable parameters, which is 4.98 MB or approximately 0.13 percent of the backbone. The preprocessor establishes a sequence length ceiling of 512 tokens to accommodate full 3-turn dialogue histories. Fine-tuning is conducted with a batch size of 1 over 3 epochs using the AdamW optimiser with a learning rate of 1e-4 and weight decay of 0.01. Vectorised prediction generation across partitions uses an inference batch size of 8. Additionally, a lightweight TF-IDF and token fuzzy-matching clinical engine operates alongside the model to normalise local dialect expressions, informal pain descriptions, and regional suburb locations onto canonical body regions and scheme triage categories.

5. Evaluation

How will you measure it?

Task 1 conversational triage is evaluated using open-ended Natural Language Generation metrics including ROUGE-1 for unigram overlap, ROUGE-2 for bigram overlap, ROUGE-L for longest common sequence, BLEU-2 for precision with smoothing, and Semantic Embedding Cosine Similarity using SentenceTransformer all-MiniLM-L6-v2. Model robustness is also evaluated through a dedicated Prompt Sensitivity Analysis across four South African greeting registers, namely isiZulu, isiXhosa, Urban Slang, and Formal English. Task 2 claim policy classification is evaluated as a multi-class closed-set classification problem using overall Accuracy, Weighted Precision, Weighted Recall, and Category-Specific F1-Scores across the 235 training, 29 validation, and 30 unseen test partitions.

What does success look like?

Task 1 conversational triage succeeds if all emergency presentations are accurately identified and prioritized for casualty routing, even when prefixed with multilingual greetings like Molo dokotela or Sawubona. Task 2 policy classification succeeds by eliminating false approvals on out-of-cover claims, achieving high accuracy across coverage categories, and ensuring administrative guidance remains safe, reliable, and free from misleading routing predictions.

6. Impact & stakeholders

Main benefit

As an example, a vulnerable young woman who hesitates to discuss sensitive symptoms with traditional elders can privately describe her condition in comfortable local phrasing, gaining immediate non-judgemental triage guidance and clear statutory benefit explanations.Generally, the primary benefit of Care2ClaimZA-LLM is empowering South African medical scheme members to navigate complex healthcare coverage with transparency and confidence. By converting informal layperson symptom descriptions into plain-language triage guidance and classifying claims into clear administrative categories, the system demystifies dense policy rulebooks. Consequently, patients can verify their entitlement to statutory Prescribed Minimum Benefits and network provider coverage upfront. This prevents unexpected out-of-pocket medical debt and reduces health anxiety during medical crises. Furthermore, the system assists healthcare providers by streamlining patient intake procedures, eliminating administrative bottlenecks, and enabling patients to articulate their symptoms effectively during consultations.

Main risk

By continuing the example, if the system misinterprets the young woman's informal description or misclassifies an urgent condition, she risks delaying essential hospital treatment or incurring unexpected out-of-pocket medical debt. Generally, the primary risk stems from potential model misclassifications and hallucinated guidance. Inaccurate coverage determinations could lead patients to incur unexpected out-of-pocket medical debt or delay necessary clinical interventions through unsafe self-triage. Additionally, relying on synthetic training data and local vernacular prompts creates a risk of encoded linguistic bias. Consequently, the model may exhibit higher error rates for underrepresented regional dialects or individuals with lower digital literacy. Over-reliance on automated decision-making also risks deflecting vulnerable patients away from

appropriate care. To mitigate these hazards, the framework strictly prohibits autonomous clinical diagnoses and incorporates human clinician oversight for complex administrative decisions.

Stakeholders

Group	What matters to them	Risks they face	Power	How we'll involve them
Individual healthcare consumers, medical scheme members, and patients navigating private healthcare funding and statutory Prescribed Minimum Benefits across South Africa	Transparent benefit explanations, predictable out-of-pocket expenses, empathetic non-diagnostic triage guidance, and strict data privacy protection under the Protection of Personal Information Act	Unexpected out-of-pocket medical debt due to misaligned provider networks, heightened health anxiety from inaccurate self-diagnosis engines, and administrative claim rejections arising from model misclassifications	Low	By testing plain-language conversational intake models directly with users, integrating regional idiom feedback, and establishing patient advocates on the regional advisory board to co-validate system updates
General practitioners, emergency medical personnel, clinic administrative intake staff, and clinical adjudicators operating within local care networks	Streamlined clinical intake procedures, reduced pre-authorisation bottlenecks, accurate patient symptom articulation during consultations, and the elimination of unvetted AI diagnostic claims	Operational administrative overload caused by misinformed patient expectations, potential clinical liability if patients delay seeking acute care based on erroneous automated advice, and workflow disruptions from unverified pre-authorisation queries	Medium	By enforcing mandatory human clinician sign-off protocols for acute triage cases, involving clinical adjudicators in benchmark evaluations, and conducting regular operational audits with clinic intake personnel.
Local patient advocates, civil society representatives, municipal energy and	Environmental sustainability, resource efficiency during peak electrical grid shortages, equitable healthcare	Local power grid overload or municipal water depletion from energy-intensive cloud data center	High	By establishing a formal Community and Healthcare Advisory Board with decision-making authority over

Group	What matters to them	Risks they face	Power	How we'll involve them
water resource managers, and public health officials	access across diverse socio-linguistic groups, and institutional algorithmic accountability	compute, alongside systemic deflection of low-income patients toward overburdened public health facilities.		dynamic compute throttling protocols, auditing regional deployment impacts, and empowering members to trigger automated fallback circuit breakers when municipal resources face critical stress

7. Governance & sustainability

How will you reduce your main risk?

To mitigate core clinical, financial, and privacy risks in this proof of concept, Care2ClaimZA-LLM implements realistic technical and operational safeguards. Firstly, the system maintains strict scope boundaries. It acts exclusively as a non-diagnostic administrative tool while requiring human clinician escalation for acute triage cases. Secondly, privacy risk is managed by using de-identified synthetic training data based on public regulatory frameworks. This ensures compliance with the Protection of Personal Information Act without exposing private health records. Thirdly, short-term technical controls address classification errors. Applying zero-temperature sampling and constrained logit decoding restricts outputs strictly to valid policy categories, eliminating preambles. Finally, explicit disclaimers clarify that outputs provide administrative guidance rather than formal scheme guarantees.

Which governance approaches will you use?

Follow existing laws (e.g. data protection), Set clear rules on uses that are not allowed, Test, monitor and add safeguards, Non-Binding Administrative Disclaimers: Every user interaction includes explicit disclaimers stating that responses provide non-diagnostic administrative guidance rather than formal scheme guarantees or medical advice.

Describe your governance plan

The governance plan enforces legal compliance, strict operational boundaries, and technical safeguards. Firstly, it ensures compliance with South Africa's Protection of Personal Information Act by using de-identified datasets and synthetic prompts without accessing private medical records. Secondly, bright line rules prohibit autonomous clinical diagnoses, unvetted care denials, and direct health record access. Acute clinical cases mandate immediate human clinician escalation. Thirdly, technical safeguards like zero-temperature sampling restrict outputs strictly to valid decision categories. Finally, scheme adjudication in this proof of concept relies on a simplified, synthetically generated dataset to enrich public datasets for LLM instruction-tuning. These mock categories have no legal bearing on actual insurance decisions. Production deployment will require indexing official policy documents from a representative set of medical aids.

Who is responsible if something goes wrong?

Responsibility is shared across key stakeholders. The developers bear primary liability for system operations, data privacy under POPIA, and algorithmic integrity. Prior to any production deployment, clinical adjudicators and intake personnel retain full responsibility for verifying triage guidance and making final coverage determinations. Meanwhile, patients remain responsible for describing symptoms accurately and seeking professional medical care when advised.

How will you reduce environmental impact?

Environmental impact is minimised by leveraging parameter-efficient techniques rather than training models from scratch. Fine-tuning Gemma 3 1B using Low-Rank Adaptation (LoRA) significantly reduces compute energy and data centre emissions. Additionally, integrating a lightweight sentence transformer for local language matching keeps overall model complexity low. This streamlined footprint enables efficient deployment on edge devices operating under low bandwidth and poor internet coverage across urban and rural South Africa.